

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and implemented **System Verilog Assertions (SVA)** and functional constraints for complex digital design blocks within an **AI accelerator**, targeting pre-silicon verification.
- Designed and built **System Verilog testbenches**, including random test generators and high-level transactional models, to validate NPU IP components.
- Collaborated with **AI architects** and design engineers to define verification strategies and develop test plans for next-generation **NPU AI accelerators**.
- Authored and executed comprehensive test cases, focusing on functional correctness and performance analysis of digital design blocks.
- Integrated **Python scripting** into the verification flow for automating test case generation and results analysis, processing approximately 150 test runs daily.
- Managed **CI/CD pipelines** using Jenkins for continuous integration and deployment of verification environments, reducing build times by **18%**.
- Participated in **Agile ceremonies** and conducted code reviews for verification code, ensuring adherence to best practices and team standards.

SKILLS

Verification Languages – System Verilog, System Verilog Assertions (SVA)

Verification Methodologies – Formal Verification, Assertion-Based Verification, UVM Simulation

AI Hardware – NPU IP, AI Accelerators

Programming & Scripting – Python, C, C++, Shell Scripting, TCL

Tools & Protocols – ASIC, SoC, FPGA, AMBA, NoC, Jenkins, Git

PROJECTS

Pre-Silicon Verification for AI Accelerator IP

- Developed a comprehensive **System Verilog** verification environment for a critical component of an **AI accelerator**, integrating **SVA properties** and functional constraints.
- Implemented **transactional models** and checkers to simulate complex data flows and validate the functional correctness of the IP block under various stress conditions.
- Generated detailed coverage reports and analyzed results using **Python scripts** to identify gaps in verification and guide further test development.

Formal Verification Environment for NPU Core

- Applied **formal verification techniques** to prove the absence of critical bugs in the NPU core's control logic, analyzing approximately 5,000 assertions.
- Developed reusable **formal constraints** and properties in System Verilog to enhance the efficiency and coverage of the formal verification process.
- Collaborated with design engineers to interpret formal analysis results and refine design specifications, ensuring adherence to architectural requirements.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)